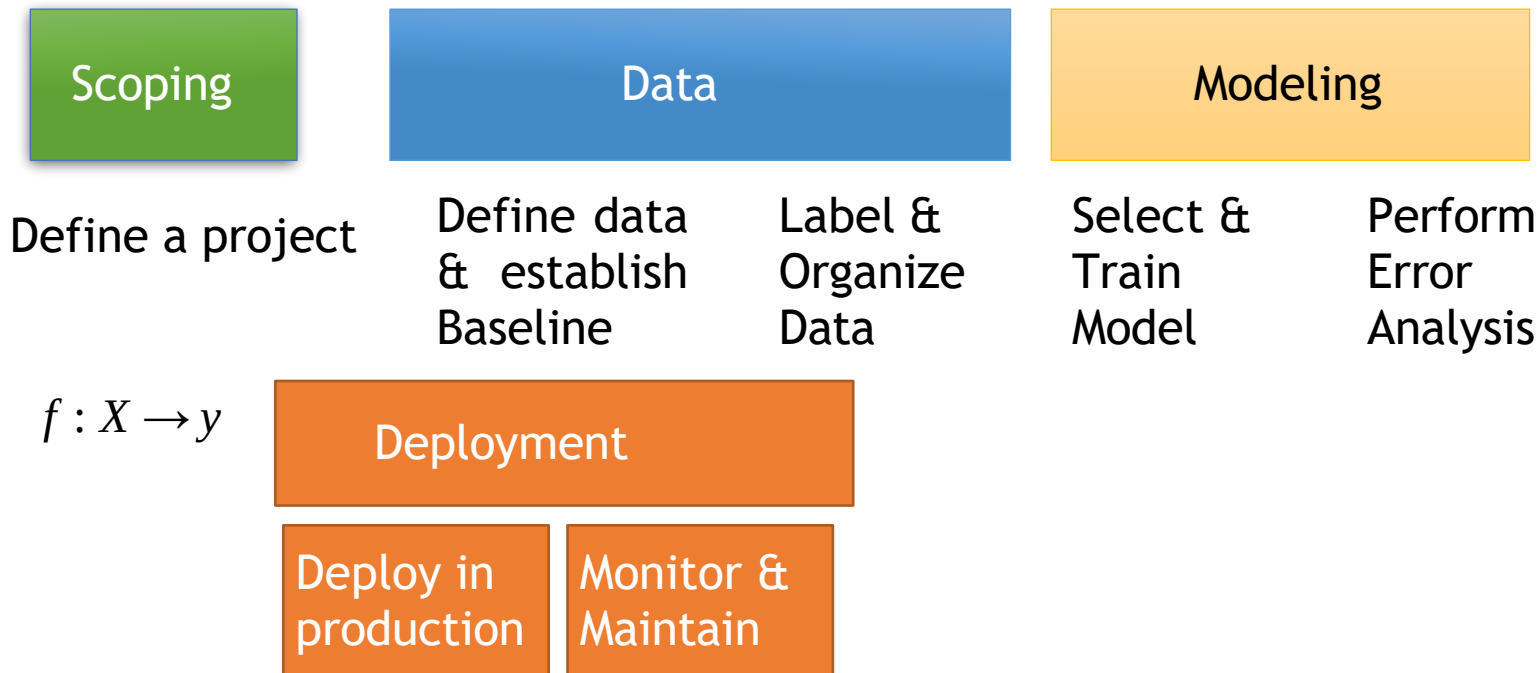
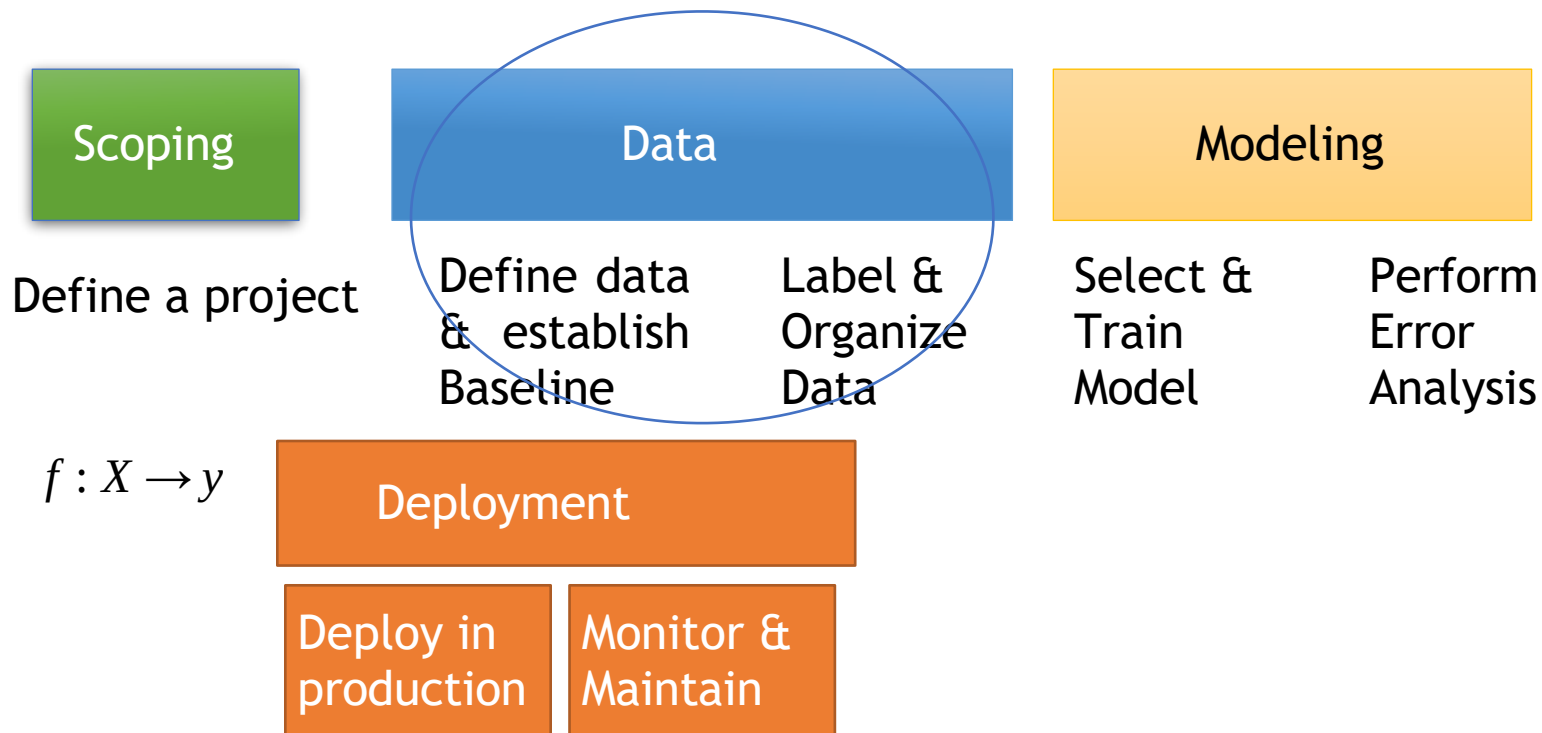

MSML 605

Data Labeling

ML Project: lifecycle



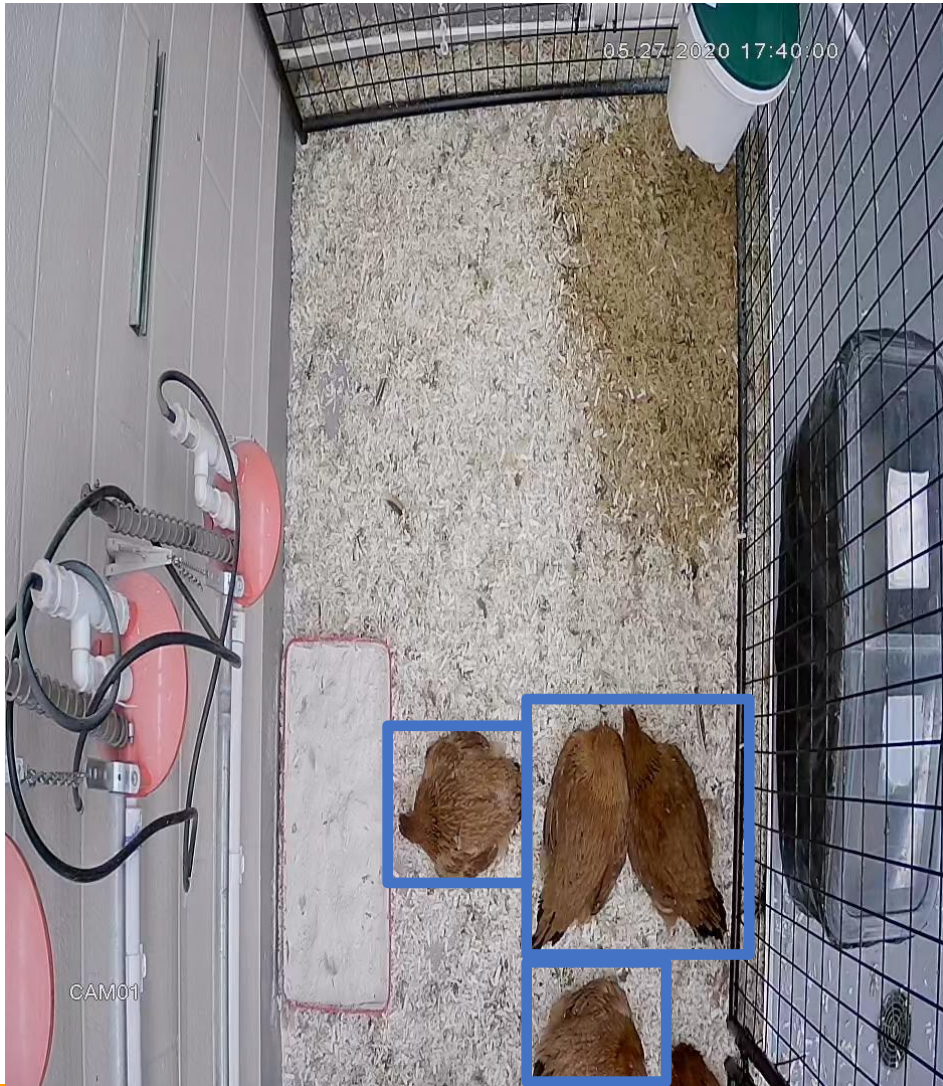
ML Project: lifecycle



Data labeling



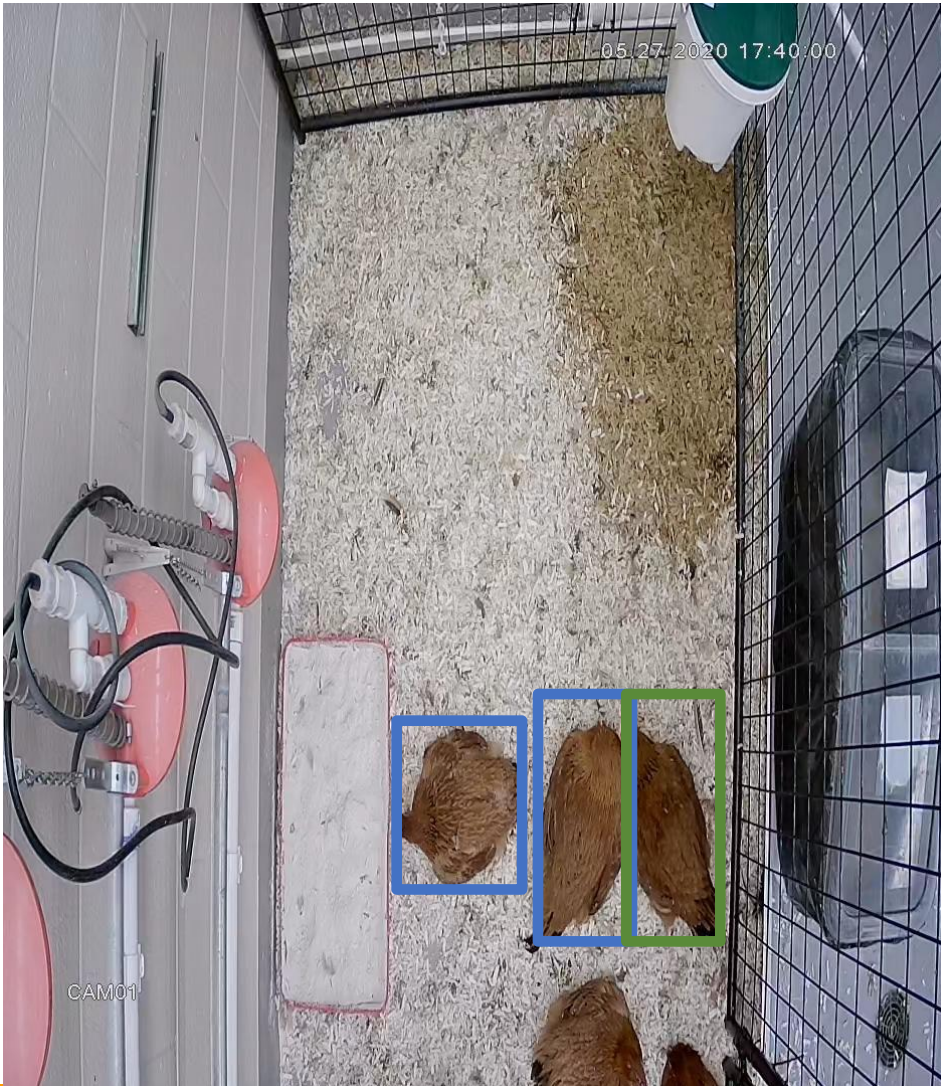
Data labeling



Data labeling



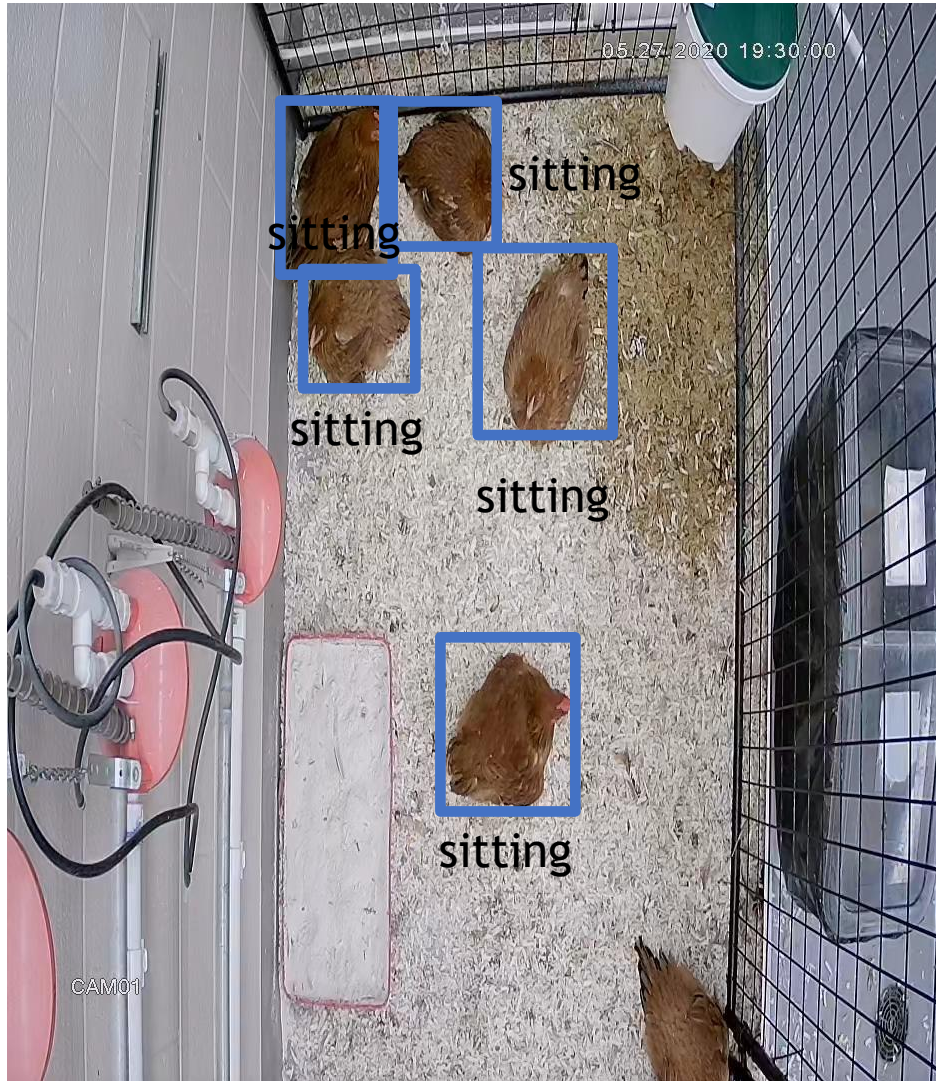
Data labeling



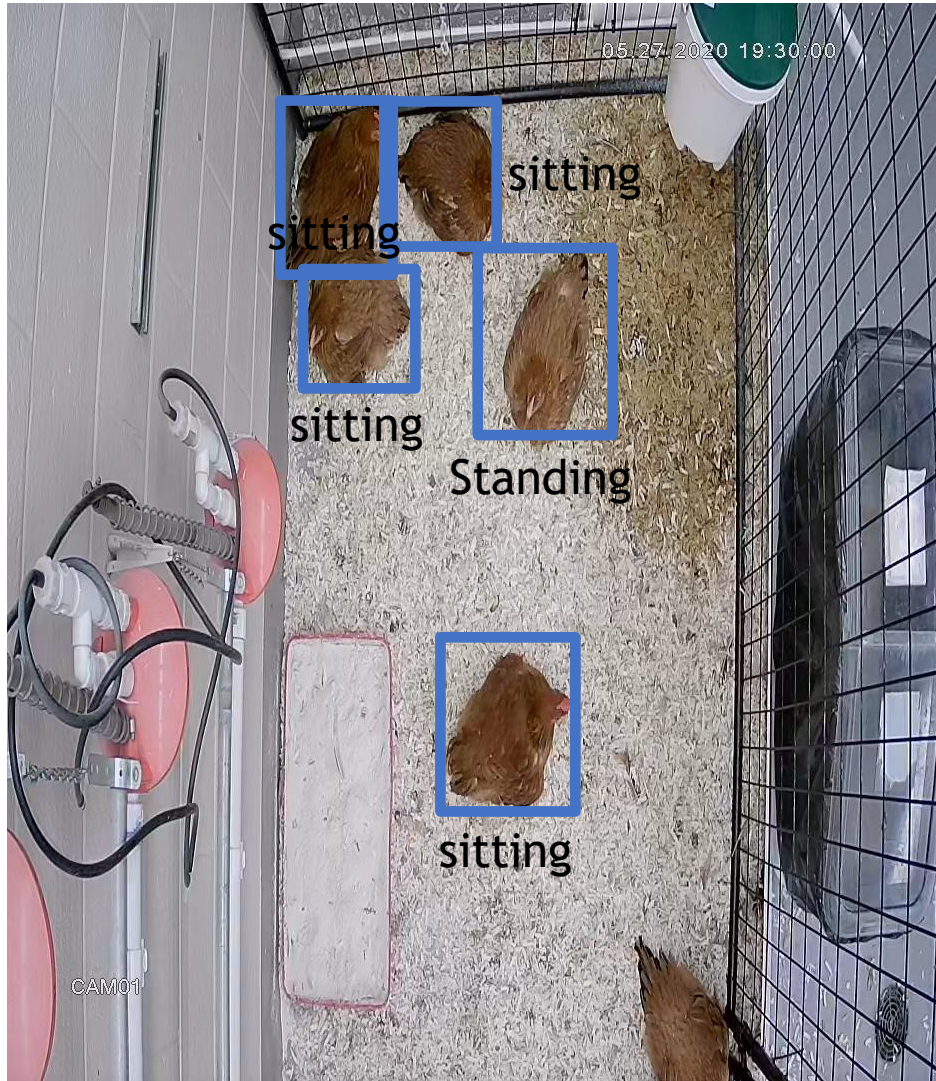
Data labeling



Model labeling



Model labeling



Sentiment analysis

- **Subjectivity Interpretation:**

Review 1: "The product is okay, but it could be better."

Labeler A might interpret this as "Neutral" because the review mentions both positive and negative aspects.

Labeler B might interpret it as "Negative" because of the phrase "could be better," indicating dissatisfaction.

- **Contextual Understanding:**

Review 2: "I love this product! It's the best thing I've ever bought!"

Labeler A might label this as "Positive" because of the enthusiastic language.

Labeler B might label it as "Neutral" if they interpret the review as potentially biased or exaggerated.

- **Language Ambiguity:**

Review 3: "The product was not bad."

Labeler A might label this as "Neutral" because it's not explicitly positive or negative.

Labeler B might label it as "Negative" because of the negative connotation of "not bad," indicating dissatisfaction.

Sentiment analysis

- **Cultural or Demographic Differences:**

Review 4: "The product is fine."

Labeler A might label this as "Neutral" because "fine" could imply a neutral sentiment.

Labeler B, who might belong to a different demographic or culture, might label it as "Positive" because in their context, "fine" might indicate satisfaction.

- **Inconsistencies in Guidelines:**

The guidelines provided to labelers might be open to interpretation or ambiguous, leading different labelers to apply them differently.

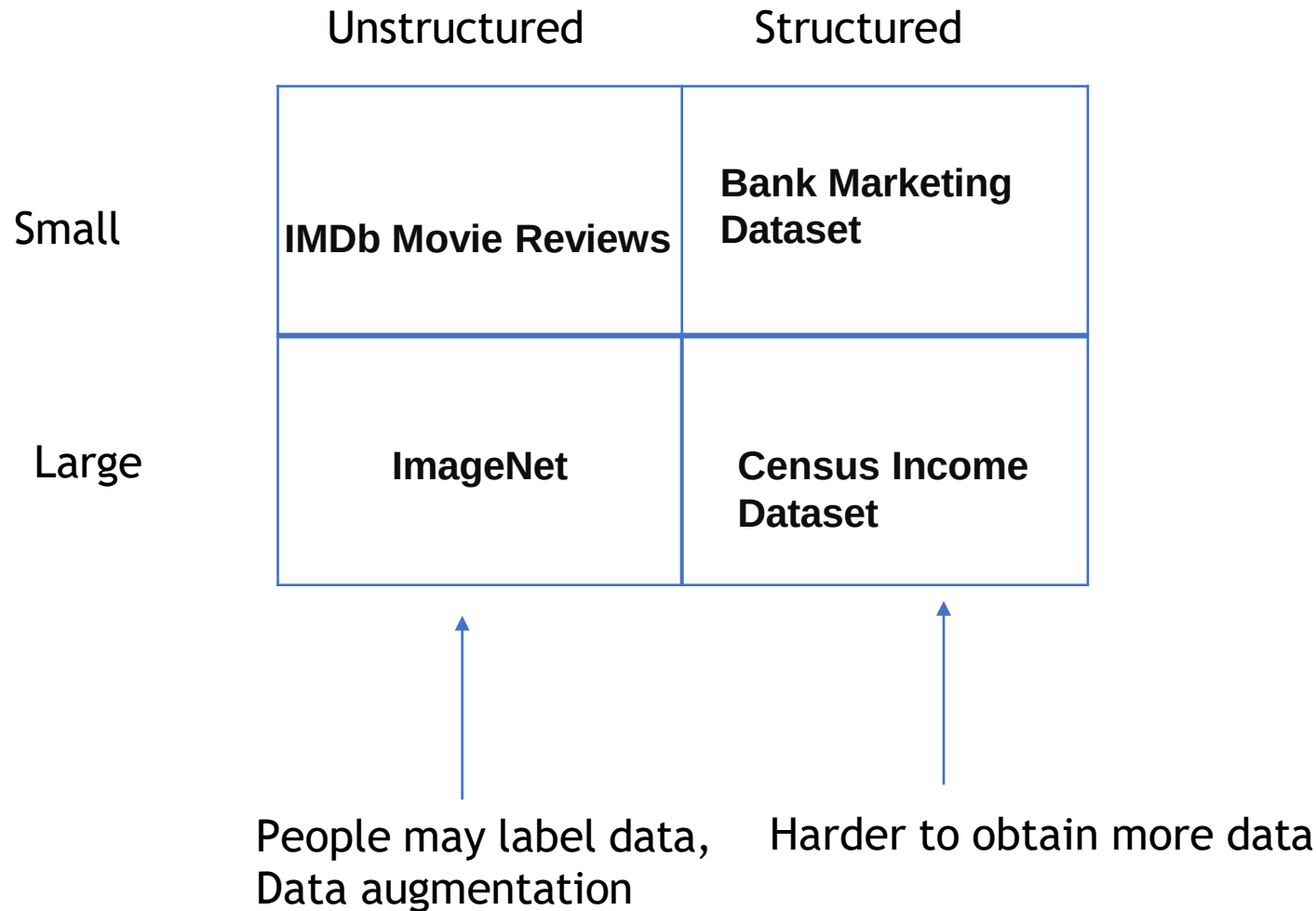
Structured vs unstructured data

Structured	Unstructured
Data with a predefined format and organization	Data lacking a predefined format or structure
Typically stored in tables or databases	May be stored in various formats (text, images, audio, video, etc.)
Databases, spreadsheets, CSV files	Text documents, images, audio recordings, videos
Often straightforward, with clear fields and categories	Often requires manual annotation or tagging, can be subjective
Dealing with missing values, handling categorical variables, ensuring data consistency	Subjectivity in annotation, dealing with ambiguity and context, scalability
Easier to automate labeling processes using rules and algorithms	Challenging to automate, often requires human judgment and expertise
Well-suited for structured queries and statistical analysis	Requires specialized techniques for processing and analysis, such as natural language processing (NLP) for text data, computer vision for image data, etc.
Structured data can be easier to scale and manage as it grows	Unstructured data may require more storage and computational resources as it grows
Data is often interpreted based on predefined schema or attributes	Interpretation may vary based on context and subjective understanding
Customer databases, financial records, inventory management	Social media analysis, sentiment analysis, image classification, speech recognition

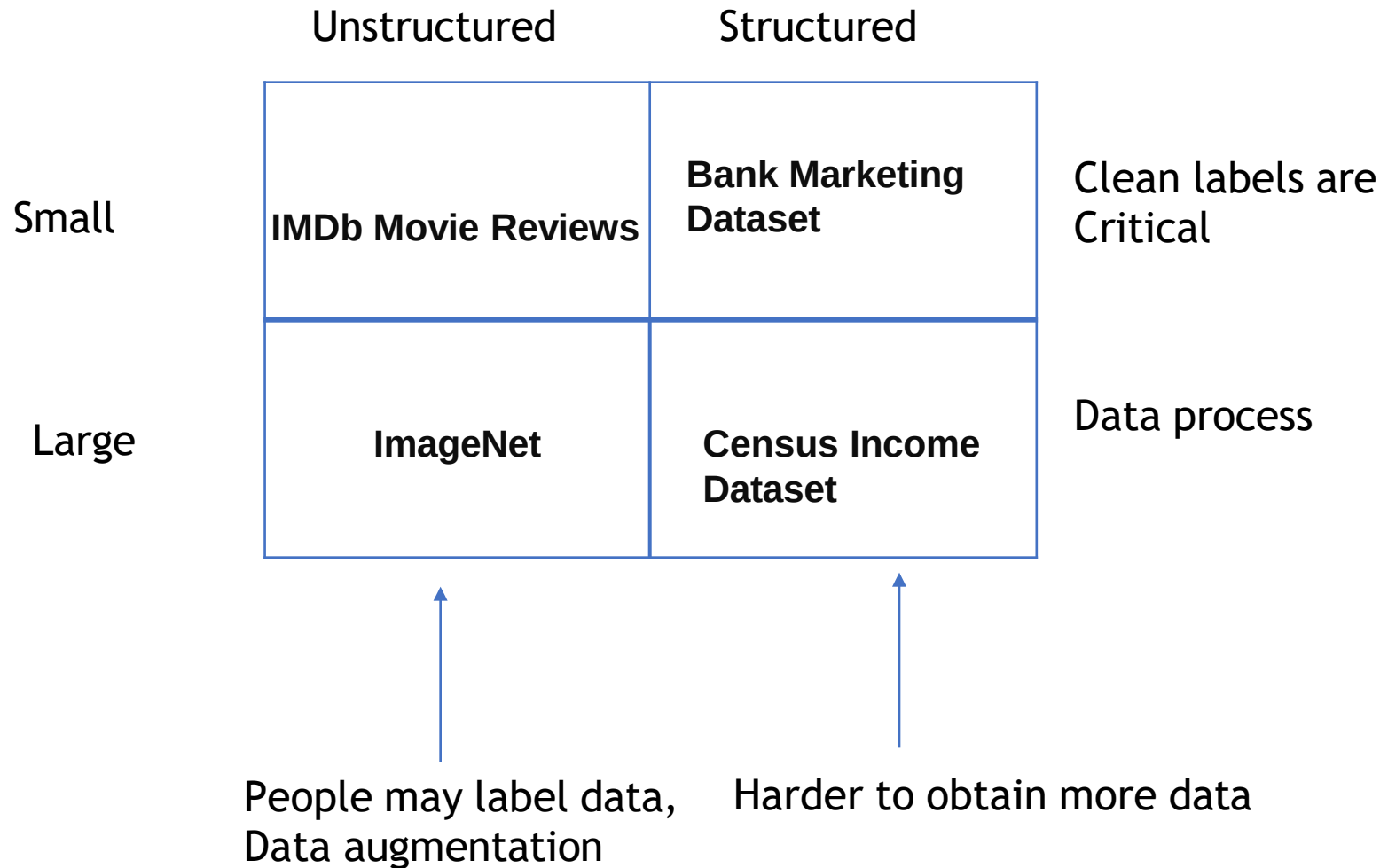
Structured & unstructured dataset

	Unstructured	Structured
Small	IMDb Movie Reviews	Bank Marketing Dataset
Large	ImageNet	Census Income Dataset

Structured & unstructured dataset



Structured & unstructured dataset



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Small vs large data

Small	Large
Limited number of examples, typically manageable in memory	Large volume of examples, potentially requiring distributed storage and processing
Labeling process may be less time-consuming and labor-intensive	Labeling process can be more time-consuming and labor-intensive, requiring more resources
Lower labeling cost due to fewer examples to annotate	Higher labeling cost due to the larger number of examples requiring annotation
Easier to maintain labeling consistency across a smaller dataset	Ensuring labeling consistency may be more challenging due to the larger dataset size
Potential for higher quality labels with closer attention to detail	Quality control becomes crucial due to the scale of the dataset, may require automated checks and human review
Basic annotation tools may suffice for labeling small datasets	Advanced annotation tools or crowdsourcing platforms may be necessary for labeling large datasets efficiently
Limited potential for automation due to smaller dataset size	Automation through machine learning or crowdsourcing becomes more viable as dataset size increases
Requires less expertise for labeling, can be done by a smaller team	May require more specialized expertise and larger annotation teams
Easier to explore and analyze small datasets using traditional tools and techniques	Requires specialized tools and techniques for analyzing large datasets, such as distributed computing frameworks
Less emphasis on scalability, as the dataset size is inherently small	Scalability becomes a significant concern, requiring infrastructure and systems capable of handling large volumes of data

Small data challenges in big data

- **Data Quality and Completeness:**

individual records may still suffer from issues such as missing values, duplication, or inaccuracies

These issues can propagate across the entire dataset, leading to data quality and completeness challenges similar to those encountered in small datasets.

- **Sample Bias:**

In a big data scenario, where data is collected from various sources and channels, sample bias can still occur.

if a particular demographic is overrepresented in the data due to sampling biases or selection processes,
the analysis results may not generalize well to the broader population

Small data challenges in big data

- **Scalability of Algorithms:**

Algorithms that perform well on small datasets may not scale efficiently to big data due to computational limitations or memory constraints

an algorithm that has quadratic time complexity may become impractical to run on massive datasets

- **Rare events:**

For example, in self-driving cars.

Improving labelling

- **Clear Labeling Guidelines**
- **Training and Familiarization**
- **Inter-labeler Agreement**

Calculate metrics such as Cohen's kappa or Fleiss' kappa to quantify agreement levels.

- **Iterative Labeling Process**
 - **Quality Assurance Checks**
 - **Adjudication Mechanisms**
 - **Feedback and Calibration**
 - **Use Consensus Labeling**
 - **Continuous Monitoring and Feedback Loop**
-

Cohen's Kappa

- Statistic is calculated using the observed agreement between raters and the agreement that would be expected by chance.

$$K = \frac{P_o - P_e}{1 - P_e}$$

P_o - proportion of observed agreement between raters

P_e - proportion of agreement expected by chance

Cohen's Kappa

- Cohen's Kappa ranges from -1 to 1
- $K = 1$ - perfect agreement between raters
- $K = 0$ - agreement equal to what would be expected by chance alone
- $K < 0$ - agreement worse than what would be expected by chance
- $K > 0$ - agreement better than what would be expected by chance
- $K = -1$ - total disagreement

Cohen's Kappa Interpretation

- $\kappa < 0.20$ poor agreement
- $0.21 \leq \kappa \leq 0.4$ fair agreement
- $0.41 \leq \kappa \leq 0.60$ moderate agreement
- $0.61 \leq \kappa \leq 0.80$ substantial agreement
- $\kappa > 0.80$ almost perfect agreement

Cohen's Kappa Interpretation

		Rater B	
		Positive	Negative
Rater A	Positive	25	5
	Negative	10	10

$p_o = (\text{Both said Yes} + \text{Both said No}) / (\text{Total Ratings})$

$$p_o = (25 + 10) / (50) = 0.7$$

$$P(\text{"Positive"}) = ((25+5)/50) * ((25+10)/50) = 0.42$$

$$P(\text{"Negative"}) = ((10+10)/50) * ((5+10)/150) = 0.12$$

$$p_e = 0.42 + 0.12 = 0.54$$

$$k = (p_o - p_e) / (1 - p_e) \quad k = (0.7 - 0.54) / (1 - 0.54) \quad k = 0.348 \text{ Fair agreement}$$

Merging labels

Product Subcategories:

Original Labels: "Smartphones", "Tablets", "Laptops", "Desktops", "Wearable Devices"

Merged Labels: "Mobile Devices", "Computers", "Wearable Devices"

Vehicle Types:

Original Labels: "Sedan", "SUV", "Hatchback", "Crossover", "Minivan"

Merged Labels: "Cars", "SUVs", "Minivans"

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Merging labels

Music Genres:

Original Labels: "Rock", "Pop", "Hip-hop", "Jazz", "Electronic"

Merged Labels: "Popular Music", "Experimental Music"

Hotel Ratings:

Original Labels: "1 Star", "2 Stars", "3 Stars", "4 Stars", "5 Stars"

Merged Labels: "Budget Accommodation (1-2 Stars)", "Standard Accommodation (3 Stars)", "Upscale Accommodation (4-5 Stars)"

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Ambiguity in labeling

- In a dataset containing social media posts labeled as "positive", "negative", or "neutral" sentiment, some posts may contain ambiguous language or sentiment.
 - sarcasm, humor, or slang expressions can make it challenging to accurately categorize the sentiment
- In a dataset containing customer reviews with star ratings (e.g., 1-star, 2-star, 3-star), the assignment of ratings might be ambiguous for borderline cases.
 - For example, determining whether a review warrants a 3-star or 4-star rating can be subjective and vary depending on individual interpretations.
- In a dataset used for facial expression recognition, assigning labels such as "happy", "sad", "angry", etc., to facial images can be ambiguous, for mixed emotions.
 - Annotators may have different thresholds for categorizing expressions, leading to inconsistency in labeling.

Ambiguity in labeling

- In a dataset where movies are labeled with genres (e.g., "action", "comedy", "drama"), some movies may exhibit characteristics of multiple genres, making genre classification ambiguous.
 - a movie containing elements of both action and comedy may be challenging to categorize into a single genre.
 - In a dataset categorizing products into different categories (e.g., "electronics", "apparel", "home goods"), certain products may blur the lines between categories or belong to multiple categories simultaneously.
 - a smartwatch with fitness tracking capabilities may fall under both "electronics" and "health & fitness" categories, leading to ambiguity in classification.
 - In a dataset where customer support tickets are automatically tagged with issue categories (e.g., "technical", "billing", "product feature request"), the automated tagging process may misclassify tickets with ambiguous or vague descriptions.
 - a ticket mentioning "slow performance" could be categorized as either "technical" or "product feature request", depending on interpretation.
-

Human level performance - when to use

- **Benchmarking**

- By comparing model performance to human performance, we can assess the capabilities and limitations of the models relative to human intelligence.

- **Goal Setting**

- Human-level performance can serve as a target or goal for machine learning projects.

- **Quality Assurance:**

- If a model performs close to or better than humans, it instills confidence in its ability to perform tasks accurately and effectively.

- **Ethical Considerations**

- Understanding human-level performance can inform ethical considerations in machine learning, especially in domains where human judgment or expertise is critical.
-

Human level performance - when not to use

- **Feasibility**

- In some cases, achieving human-level performance may be impractical or infeasible due to resource constraints, time limitations, or the complexity of the task.

- **Task Suitability**

- For tasks where humans are not the best performers or where human judgment is subjective or unreliable, using human-level performance as a benchmark may be inappropriate.

- **Domain Specificity**

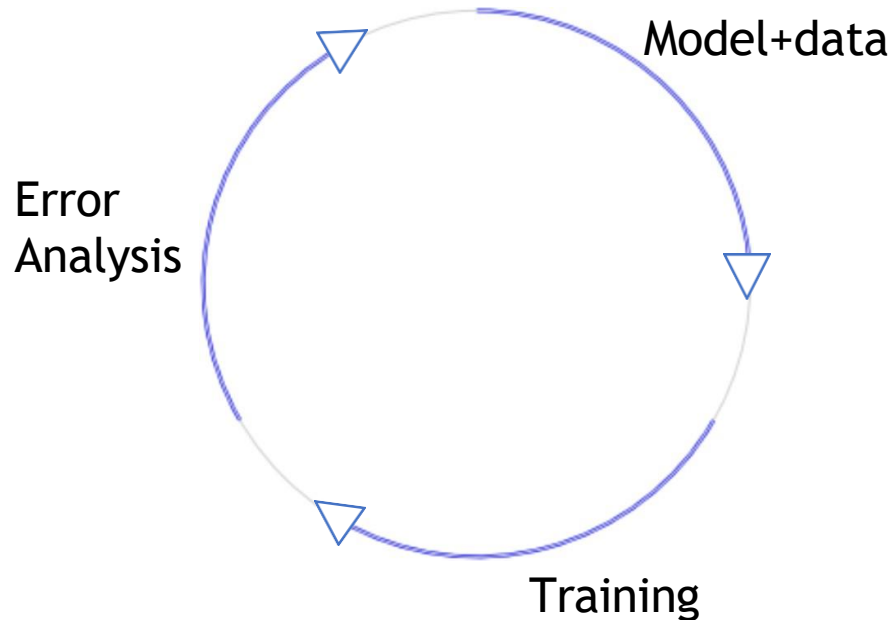
- Comparing model performance to human-level performance requires domain-specific expertise and understanding to ensure meaningful and relevant evaluations.

- **Bias and Variability**

- Using human-level performance as a benchmark may introduce bias or variability into the evaluation process, especially if human judgments are inconsistent or unreliable.
-

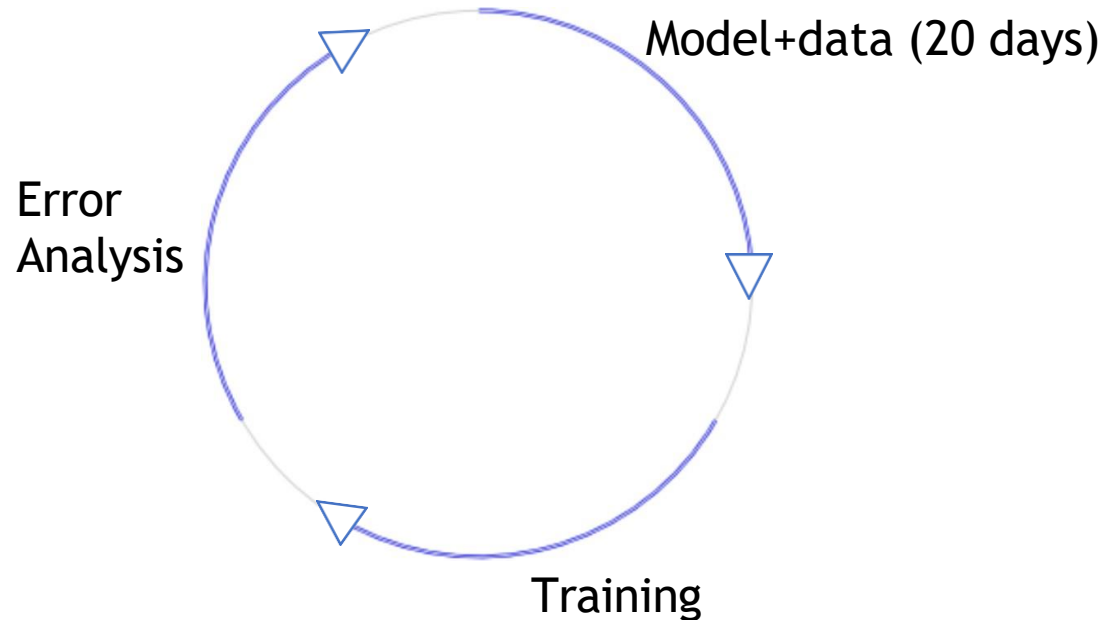
Data Collection

- Build a model, training it and error analysis pipeline



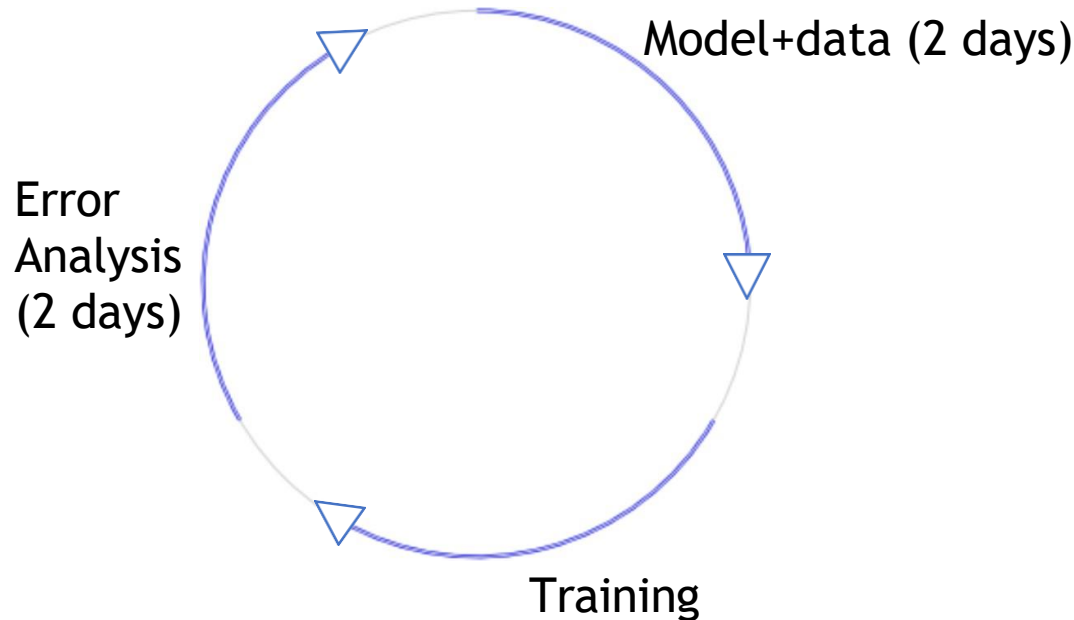
Data Collection

- Build a model, training it and error analysis pipeline



Data Collection

- Build a model, training it and error analysis pipeline





Data Pipeline

Text Data Pipeline

- **Data Collection:** Gather text data from various sources such as websites, social media, or documents.
- **Text Preprocessing:** Clean the text data by removing special characters, stopwords, and performing tokenization, stemming, or lemmatization.
- **Feature Extraction:** Convert text data into numerical features using techniques like TF-IDF (Term Frequency-Inverse Document Frequency) or word embeddings (Word2Vec, GloVe).
- **Data Transformation:** Normalize or scale the numerical features to ensure consistent ranges across features.
- **Data Splitting:** Split the data into training, validation, and test sets for model training and evaluation.

Image Data Pipeline

- **Data Collection:** Collect images from various sources or use pre-existing datasets like CIFAR-10, MNIST, ImageNet etc.
 - **Image Preprocessing:** Resize images to a consistent size, perform data augmentation (rotate, flip, zoom), and normalize pixel values.
 - **Feature Extraction:** Extract features using pre-trained CNN models (like VGG, ResNet) or extract image descriptors (like HOG, SIFT).
 - **Data Augmentation:** Generate additional training samples by applying transformations like rotation, scaling, and flipping to combat overfitting.
 - **Data Splitting:** Split the data into training, validation, and test sets.
-

Tabular data pipeline

- **Data Collection:** Gather structured data from databases, CSV files, or APIs.
 - **Data Cleaning:** Handle missing values, outliers, and perform data type conversions.
 - **Feature Engineering:** Create new features, perform one-hot encoding for categorical variables, and scale numerical features.
 - **Data Transformation:** Normalize or standardize numerical features to have zero mean and unit variance.
 - **Feature Selection:** Select relevant features using techniques like correlation analysis, feature importance, or domain knowledge.
 - **Data Splitting:** Split the data into training, validation, and test sets.
-

Time Series data

- **Data Collection:** Collect time-stamped data from sensors, IoT devices, or financial markets.
 - **Time Series Preprocessing:** Handle missing values, resample data if needed (e.g., downsampling, upsampling), and perform feature lagging or rolling window statistics.
 - **Feature Engineering:** Extract time-based features such as day of the week, month, hour, seasonality, and trends.
 - **Data Transformation:** Scale or normalize features, and handle categorical variables if present.
 - **Temporal Splitting:** Split the time series data into training and testing sets based on time to account for temporal dependencies.
-

Audio data pipeline

- **Data Collection:** Obtain audio samples from recordings, streaming services, or audio databases.
- **Audio Preprocessing:** Extract audio features such as MFCCs (Mel Frequency Cepstral Coefficients), spectrograms, or chroma features.
- **Feature Engineering:** Compute statistical features (mean, variance) or use domain-specific features like pitch, tempo, or energy.
- **Data Transformation:** Normalize or scale features, and handle feature dimensionality reduction if needed.
- **Data Splitting:** Split the data into training, validation, and test sets for model training and evaluation.